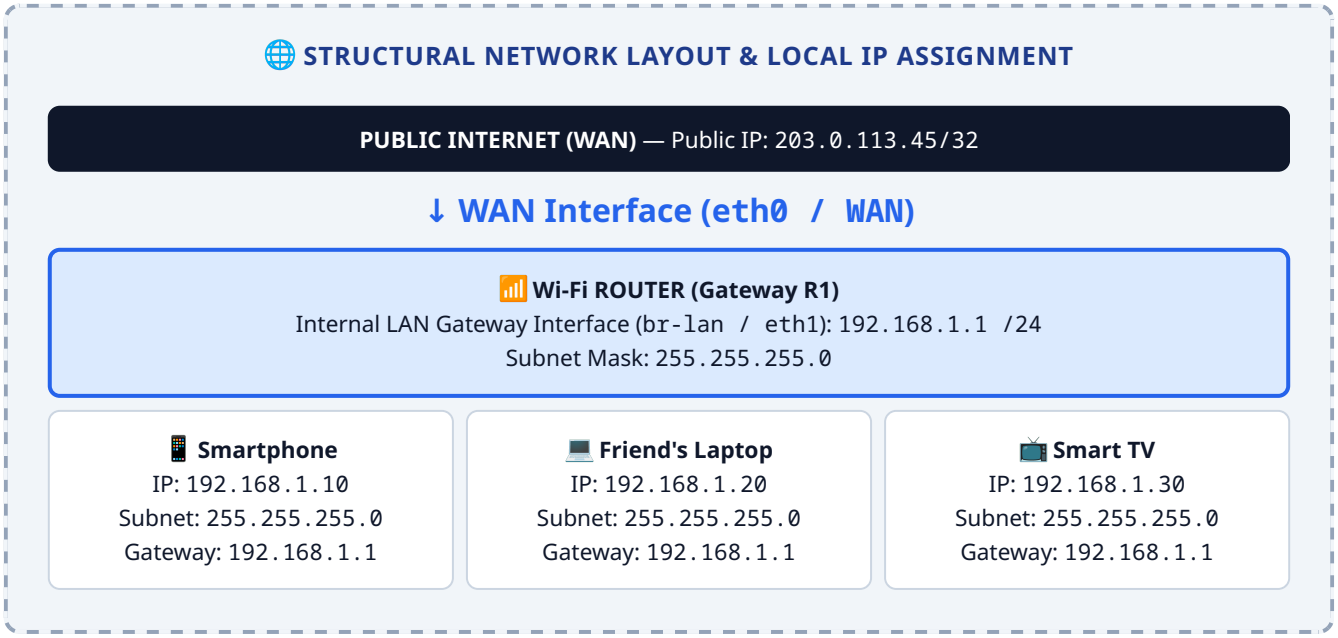
IP Routing & Network Layer Architecture Assignment

Comprehensive Long-Form Technical Report & Detailed Solutions (Tasks 1 to 5)

Task 1: Home Network IP Diagram & Interface Addressing

In a typical local area network (LAN), a Wi-Fi Broadband Router acts as the central default gateway. It routes IP traffic between internal client devices on the private LAN subnet and the external public WAN network via Network Address Translation (NAT). Below is the structural IP layout diagram showing three client devices connected through a central router:



Interface & IP Addressing Breakdown:

- **Router LAN Interface (`192.168.1.1/24`):** Serves as the internal Default Gateway for all local devices on the `192.168.1.0/24` subnet.
- **Smartphone (`192.168.1.10/24`):** Dynamically assigned via DHCP over Wi-Fi ($5 ext{ GHz}$ band).
- **Friend's Laptop (`192.168.1.20/24`):** Connected wirelessly or via Ethernet patch cable.
- **Smart TV (`192.168.1.30/24`):** Connected via RJ-45 Ethernet cable to the router's LAN port for fast media streaming.

Task 2: Step-by-Step Simulation of Inter-Subnet Packet Routing

Scenario: Your Laptop (IP: `10.1.1.50/24`) in Subnet A wants to send a data packet to a Friend's Computer (IP: `10.2.2.80/24`) in Subnet B. The two subnets are interconnected by Router R1 and Router R2 across a point-to-point link (`172.16.0.0/30`).

Network Topology Parameters:

- **Source PC (Your Laptop):** `10.1.1.50/24` | Default Gateway: `10.1.1.1`
- **Router R1:** Interface `eth0` (LAN A): `10.1.1.1/24` | Interface `eth1` (WAN Link): `172.16.0.1/30`
- **Router R2:** Interface `eth0` (WAN Link): `172.16.0.2/30` | Interface `eth1` (LAN B): `10.2.2.1/24`
- **Destination PC (Friend's Computer):** `10.2.2.80/24` | Default Gateway: `10.2.2.1`

Step-by-Step Packet Traversal & Routing Table Lookup:

Step 1: Source Host Decision (Your Laptop)

Your laptop performs a bitwise AND operation between the destination IP (`10.2.2.80`) and its own subnet mask (`255.255.255.0`). Since `10.2.2.0` does not match local subnet `10.1.1.0`, the laptop encapsulates the IP packet into an Ethernet frame with destination MAC address set to its Default Gateway (**Router R1** interface `10.1.1.1`).

Step 2: Hop 1 — Router R1 Processing & Routing Table Lookup

Router R1 receives the frame, removes the Data Link header, and inspects the IP header destination address `10.2.2.80`. R1 checks its internal routing table:

Destination Subnet	Subnet Mask	Next Hop IP	Outgoing Interface
`10.1.1.0`	`255.255.255.0`	Directly Connected	`eth0` (LAN A)
`172.16.0.0`	`255.255.255.252`	Directly Connected	`eth1` (WAN Link)
`10.2.2.0`	**`255.255.255.0`**	**`172.16.0.2`**	**`eth1`**

Routing Decision: Match found for `10.2.2.0/24`! Router R1 decrements IP TTL by 1, re-encapsulates the packet, and forwards it to Next Hop `172.16.0.2` via interface `eth1`.

Step 3: Hop 2 — Router R2 Processing & Local Delivery

Router R2 receives the packet on interface `eth0` (`172.16.0.2`) and inspects its routing table:

Destination Subnet	Subnet Mask	Next Hop IP	Outgoing Interface
`172.16.0.0`	`255.255.255.252`	Directly Connected	`eth0` (WAN Link)
`10.2.2.0`	**`255.255.255.0`**	**Directly Connected**	**`eth1` (LAN B)**

Routing Decision: Destination subnet `10.2.2.0/24` is directly connected to `eth1`! Router R2 uses ARP to resolve the MAC address of `10.2.2.80` and delivers the frame directly to your friend's computer.

Task 3: Routing Table Design for Zomato Multi-City Delivery Network

Consider a enterprise delivery network for Zomato with hub branches in three major Indian cities: **Ahmedabad Hub**, **Mumbai Hub**, and **Bengaluru Hub**. Each city branch operates on a dedicated subnet interconnected via a central MPLS / VPN core router.

Central Core Router Routing Table:

Branch Location	Network Address	Subnet Mask	CIDR	Next Hop Gateway	Outgoing Interface
Ahmedabad Hub	`10.100.10.0`	`255.255.255.0`	`/24`	`192.168.10.1`	`GigabitEthernet0/1`
Mumbai Hub	`10.100.20.0`	`255.255.255.0`	`/24`	`192.168.20.1`	`GigabitEthernet0/2`
Bengaluru Hub	`10.100.30.0`	`255.255.255.0`	`/24`	`192.168.30.1`	`GigabitEthernet0/3`
Internet Gateway (Default Route)	`0.0.0.0`	`0.0.0.0`	`/0`	`203.0.113.1`	`Serial0/0/0`

Task 4: Comparative Analysis: Static Routing vs. Dynamic Routing Protocols

Routing strategies dictate how network paths are discovered and maintained. Below is a detailed 3-Pro and 3-Con comparison between Static Routing and Dynamic Routing Protocols (such as OSPF / RIP):

Routing Type	Advantages (Pros)	Disadvantages (Cons)
Static Routing	1. **Zero CPU/RAM Overhead:** Routers do not process routing update packets. 2. **Predictable Security:** Administrators explicitly control every route. 3. **No Bandwidth Waste:** No routing protocol updates are transmitted over WAN links.	1. **High Admin Burden:** Every route must be entered manually. 2. **No Auto-Failover:** If a link dies, traffic is dropped until manual reconfiguration. 3. **Unscalable:** Extremely difficult to manage in large enterprise networks.
Dynamic Routing (OSPF / RIP)	1. **Automatic Topology Discovery:** Routers automatically detect new networks. 2. **Instant Re-Routing (Failover):** Reroutes traffic within milliseconds if a cable fails. 3. **Highly Scalable:** Easily manages hundreds of routers without manual updates.	1. **Resource Intensive:** Consumes router CPU, RAM, and link bandwidth. 2. **Configuration Complexity:** Requires skilled network engineers to configure. 3. **Potential Security Risks:** Rogue routers could inject invalid route updates if unauthenticated.

> 🍔 **Real-World App Application (How Zomato / Food Delivery Apps Benefit):**
> A food delivery platform like Zomato relies on real-world cloud microservices hosted across multiple data centers (AWS/Google Cloud). If a primary optical fiber link between the Mumbai data center and Bengaluru cloud servers gets cut, **Dynamic Routing (OSPF/BGP)** instantly re-routes restaurant search and order traffic via an alternate link in Chennai within milliseconds. If static routing were used, the entire Zomato app would crash for millions of users until an engineer manually updated the static IP routes!

Task 5: AI-Generated Cisco OSPF Configuration & Line-by-Line Explanation

Below is a sample Cisco IOS configuration for setting up **OSPF (Open Shortest Path First) Area 0** routing between two routers (Router A and Router B):

```
! --- CISCO IOS OSPF ROUTER CONFIGURATION ---
Router-A# configure terminal
Router-A(config)# router ospf 1
Router-A(config-router)# router-id 1.1.1.1
```

```
Router-A(config-router)# network 10.1.1.0 0.0.0.255 area 0
Router-A(config-router)# network 172.16.0.0 0.0.0.3 area 0
Router-A(config-router)# exit
```

Line-by-Line Technical Explanation:

- `configure terminal`: Enters Global Configuration mode on the Cisco IOS router.
- `router ospf 1`: Initializes the OSPF routing process with a local Process ID of 1.
- `router-id 1.1.1.1`: Uniquely identifies Router A within the OSPF network domain using a 32-bit dotted-decimal number.
- `network 10.1.1.0 0.0.0.255 area 0`: Enables OSPF on interfaces matching subnet `10.1.1.0/24` (using wildcard mask `0.0.0.255`) and assigns them to **Backbone Area 0**.
- `network 172.16.0.0 0.0.0.3 area 0`: Enables OSPF on the point-to-point WAN link subnet `172.16.0.0/30` (wildcard mask `0.0.0.3`) in Area 0 to establish adjacency with Router B.
- `exit`: Exits OSPF configuration mode and returns to Global Configuration mode.